IE 206: Introduction to AI & ML

Linear Regression

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Classification vs Regression

Learning setup:

- $S = \{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \ldots, (x^{(m)}, y^{(m)})\}$ is a set of data points
- $x^{(i)} \in \mathcal{X}$ for all $i = 1, 2, \ldots, m$ are drawn iid
- $y^{(i)} \in \mathcal{Y}$ is the label of $x^{(i)}$. $y^{(i)} = f(x^{(i)})$ (f is unknown)

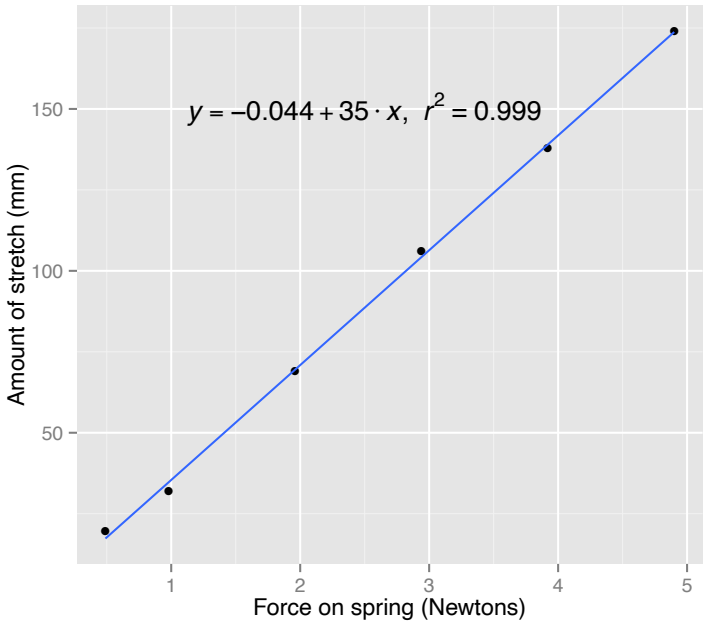
(Supervised) Learning problem: Given S (dataset) find 'best' label for a new sample $x \in \mathcal{X}$ drawn from the same distribution.

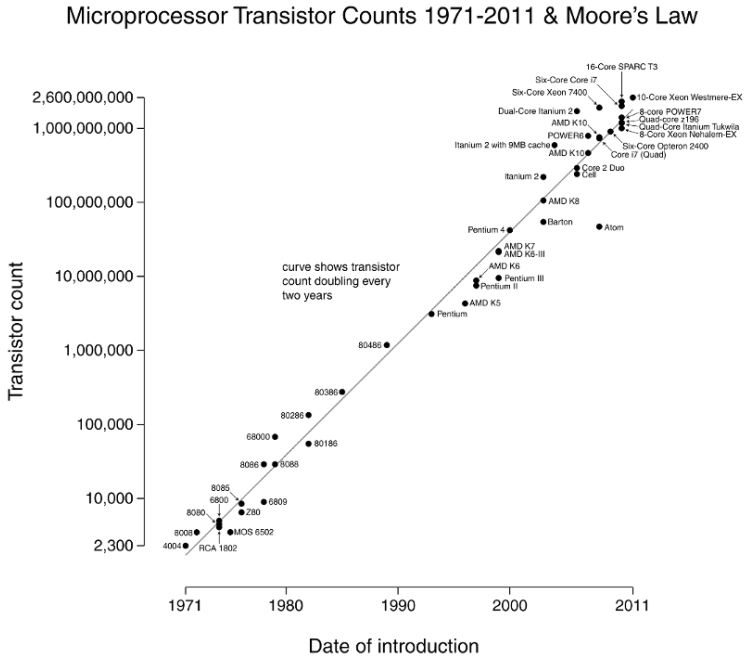
- when $|\mathcal{Y}|$ is finite, learning classification problem
- When $|\mathcal{Y}|$ is not finite/countable, learning is a regression problem

Linear Regression

We will focus on simple type of relations: **Linear**

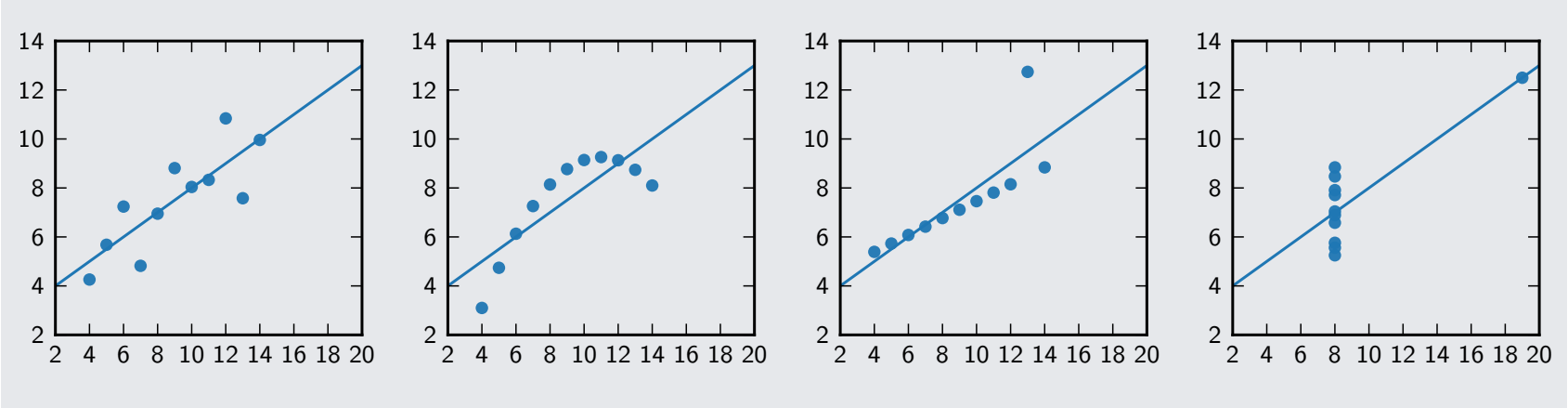
Relation between features and labels is linear.





- Amount of stretch when force is applied on a spring (spring constant!)
- Number of transistor on the silicon area doubles 18 months (Moore's Law!)

Everything is not linear!



- Data with similar quantitative (mean) may look very different
- Visualization of data is reveals patterns that are hidden in pure numerical analysis

Simple Linear Regression

- Assume: Each sample has one feature/attribute ($x^{(i)} \in \mathcal{R}$)
- We will fit line of the from $y = \beta_0 + \beta_1 x$

- x is called the independent/predictor variable
- y is called the dependent/label/response variable
- β_1 is the slope and β_0 is the intercept
- We will get different lines for different choice of (β_0, β_1)

- How to quantify how good is a line?
- Choose the best line!

Probabilistic Model for Linearly Related Data

- Instead of $y_i = \beta_1 x^{(i)} + \beta_0$ assume data is perturbed by noise
- $y_i = \beta_1 x^{(i)} + \beta_0 + \epsilon_i$, where ϵ_i is random perturbation (noise)
- perturbation denotes that data won't be fit the model perfectly
- $E[y] = \beta_0 + \beta_1 x^{(i)}$, model fits in expectation.
- We assume that $\epsilon_i \sim \mathcal{N}(0, \sigma^2)$, where σ^2 is known

Quantify goodness of a line: Mean Squared Error

- Minimize the distance between the line and points
- distance of point $(x^{(i)}, y^{(i)})$ from line (β_0, β_1) (error)

$$y^{(i)} - (\beta_1 x^{(i)} + \beta_0)$$

- As staying above or below line are equally bad we can take

$$|y^{(i)} - (\beta_1 x^{(i)} + \beta_0)| \text{ absolute error}$$

$$(y^{(i)} - (\beta_1 x^{(i)} + \beta_0))^2 \text{ squared error}$$

- We take goodness of line (β_0, β_1) as sum of the squared errors

$$\frac{1}{m}\sum_{i=1}^{m}(y^{(i)} - (\beta_1 x^{(i)} + \beta_0))^2$$

$$\min_{(\beta_0,\beta_1)} \frac{1}{m}\sum_{=1}^{n}(y^{(i)} - (\beta_1 x^{(i)} + \beta_0))^2 \text{ Mean Squared Error (MSE)}$$

Least Squared Solution

MSE is the optimal solution when error is $\sim \mathcal{N}(0,1)$.

$$(\hat{\beta}_0, \hat{\beta}_1) = \arg\min_{(\beta_0,\beta_1)} \frac{1}{m} \sum_{i=1}^{n} (y^{(i)} - (\beta_1 x^{(i)} + \beta_0))^2$$

$$\hat{\beta}_1 = \frac{\frac{1}{m}\left(\sum_{i=1}^{m} x^{(i)} y^{(i)}\right) - \left(\frac{1}{m}\sum_{i=1}^{m} x^{(i)}\right)\left(\frac{1}{m}\sum_{i=1}^{m} y^{(i)}\right)}{\frac{1}{m}\left(\sum_{i=1}^{m} (x^{(i)})^2\right) - \left(\frac{1}{m}\sum_{i=1}^{m} x^{(i)}\right)^2}$$

$$\hat{\beta}_0 = \left(\frac{1}{m}\sum_{i=1}^{m} y^{(i)}\right) - \hat{\beta}_1 \left(\frac{1}{m}\sum_{i=1}^{m} x^{(i)}\right)$$

Expressing the solutions in terms of statistics

Given a random sample $(X^{(1)}, X^{(2)}, \ldots, X^{(m)})$

- Sample mean: $\bar{X} = \frac{1}{m}\left(\sum_{i=1}^{m} X^{(i)}\right)$
- Sample variance: $S_X^2 = \frac{1}{m-1}\left(\sum_{i=1}^{m}(X^{(i)} - \bar{X})^2\right)$
- Sample standard deviations: $S_X = \sqrt{S_X^2}$.

For give data $S = \{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \ldots (x^{(m)}, y^{(m)})\}$

$$\bar{x} = \frac{1}{m}\left(\sum_{i=1}^{m} x^{(i)}\right) \qquad s_x = \frac{1}{m-1}\left(\sum_{i=1}^{m}(x^{(i)} - \bar{x})^2\right)$$

$$\bar{y} = \frac{1}{m}\left(\sum_{i=1}^{m} y^{(i)}\right) \qquad s_y = \frac{1}{m-1}\left(\sum_{i=1}^{m}(y^{(i)} - \bar{y})^2\right)$$

$$r = \frac{1}{m-1}\sum_{i=1}^{m}\left(\frac{x^{(i)} - \bar{x}}{s_x}\right)\left(\frac{y^{(i)} - \bar{y}}{s_y}\right) \quad \text{Correlation coefficient}$$

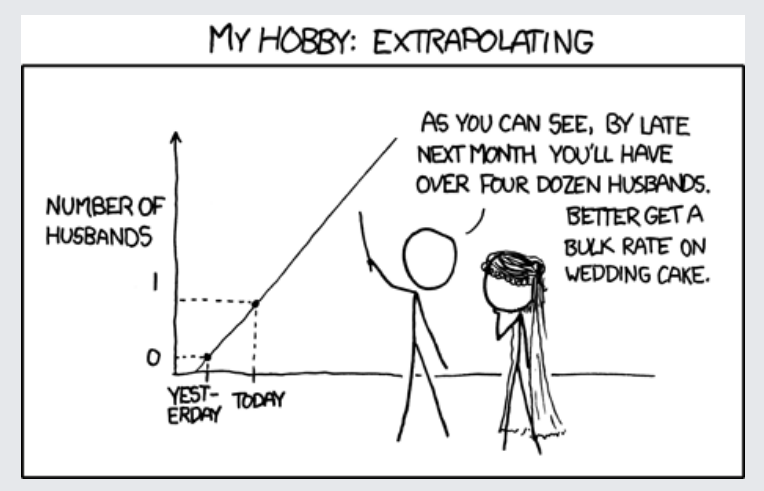
Prediction

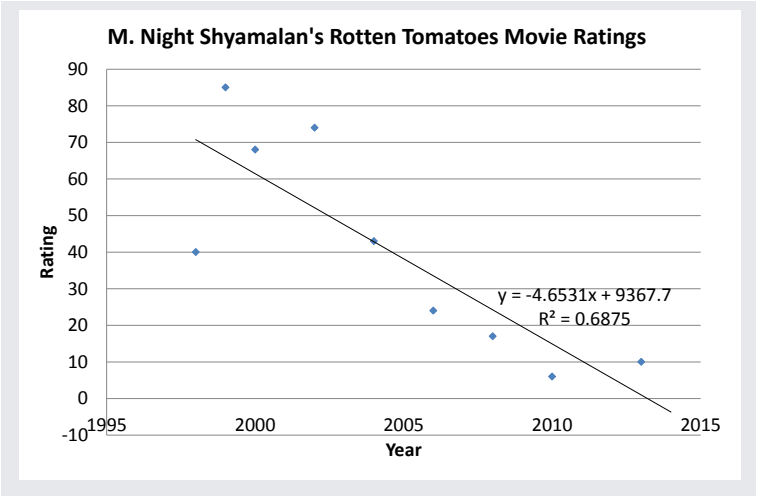
$$\hat{\beta}_1 = r\frac{s_y}{s_x} \text{ and } \hat{\beta}_0 = \bar{y} - \hat{\beta}_1\bar{x}$$

Given any sample x, its predicted label is

$$y = \hat{\beta}_1 x + \hat{\beta}_0$$

For what all x we can get prediction?

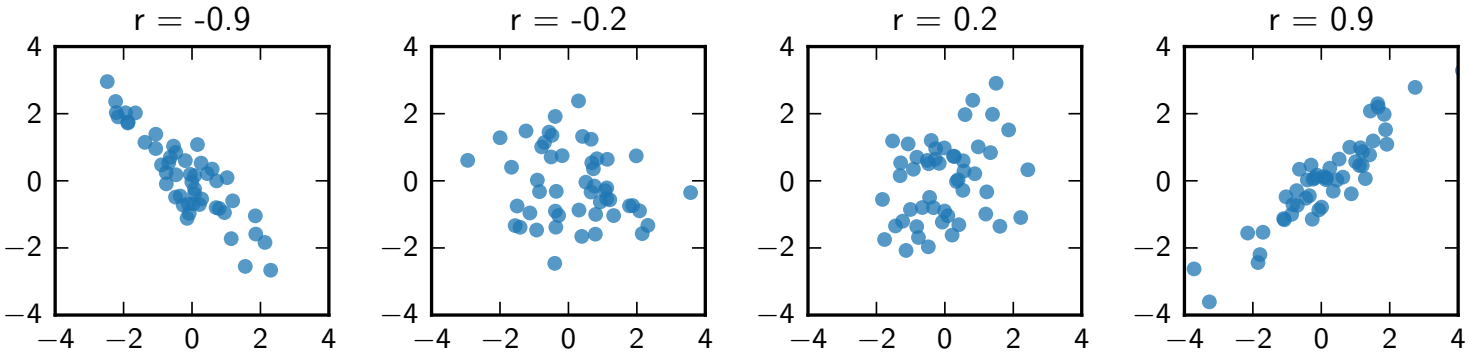




Correlation coefficient

$$r = \frac{1}{m-1} \sum_{i=1}^{m} \left(\frac{x^{(i)} - \bar{x}}{s_x} \right) \left(\frac{y^{(i)} - \bar{y}}{s_y} \right)$$

- $-1 \leq r \leq 1$. Measure how much y is related to x
- if r is positive y increases in x
- if r is negative y decreases in x



- r^2 is called coefficient of determination (explains how well data is fit).

Multiple Linear Regression

$S = \{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \ldots, (x^{(m)}, y^{(m)})\}, x^{(i)} \in \mathcal{R}^d$, $d > 1$

Each sample point $x^{(i)} = (x_1^{(i)}, x_2^{(i)}, \ldots, x_d^{(i)})$.

- We can write linear relation: $y_i = \sum_{j=1}^{d} x_j^{(i)} \beta_j + \beta_0$
- $y_i = \sum_{j=0}^{d} x_j^{(i)} \beta_j$, where $x_0^{(i)} = 1$ for all $i = 1, 2 \ldots, m$
- set $\beta^T = (\beta_0, \beta_1, \beta_2, \ldots, \beta_d)$ and $x^{(i)T} = (1, x_1^{(i)}, x_2^{(i)}, \ldots, x_d^{(i)})$
- Compactly $y_i = x^{(i)T} \beta$ for all $i = 1, 2, \ldots, m$

$$\begin{bmatrix} y^{(1)} \\ y^{(2)} \\ \vdots \\ y^{(m)} \end{bmatrix} = \begin{bmatrix} 1 & x_1^{(1)} & x_2^{(1)} & \ldots x_d^{(1)} \\ 1 & x_1^{(2)} & x_2^{(2)} & \ldots x_d^{(2)} \\ \vdots & & & \\ 1 & x_1^{(m)} & x_2^{(m)} & \ldots x_d^{(m)} \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_d \end{bmatrix}$$

$$y = X^T \beta \text{ where } X \text{ is data matrix}$$

The probabilistic model is then

$$y = X^T \beta + \epsilon \quad \text{where } \epsilon_i \quad iid \sim \mathcal{N}(0, \sigma^2)$$

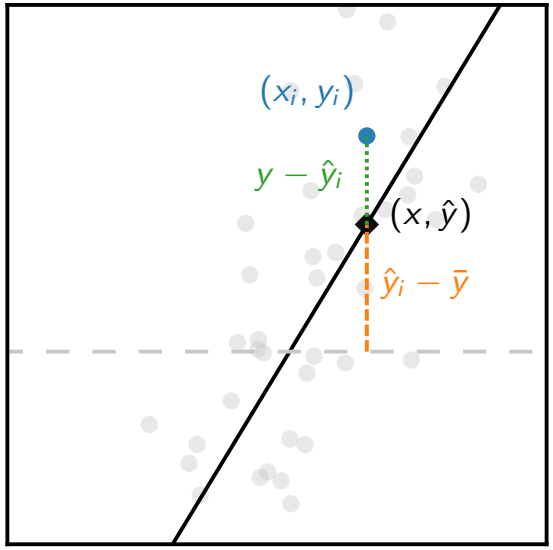
Solution of Multiple Linear Regression

$$\hat{\beta} = \arg\min_{\beta} \sum_{i=1}^{m} (y^{(i)} - x^{(i)T}\beta)^2$$

$$\hat{\beta} = (XX^T)^{-1}X^T y$$

Model Evaluation:

Suppose every point $y^{(i)}$ is very close to $\bar{y} \implies y^{(i)}$ does not dependent much on x_i and there is not much random error.

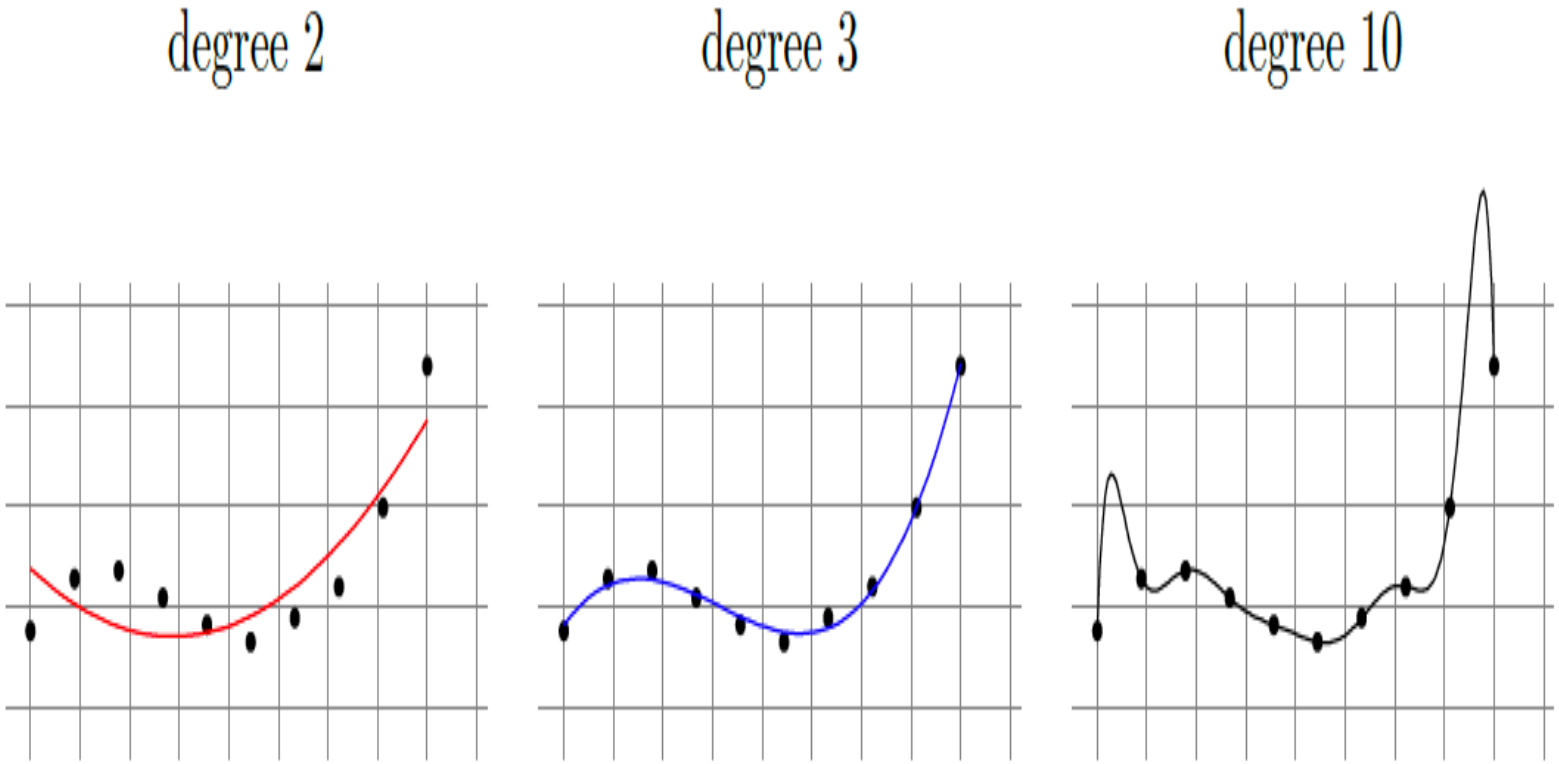


$$y^{(i)} - \bar{y} = \underbrace{(\hat{y}^{(i)} - \bar{y})}_{\text{explained by model}} + \underbrace{(y^{(i)} - \hat{y}^{(i)})}_{\text{not explained by model}}$$

Coefficient Determination

$$\underbrace{\sum_i (y^{(i)} - \bar{y})^2}_{\text{SST}} = \underbrace{\sum_i (\hat{y}^{(i)} - \bar{y})^2}_{\text{SSM}} + \underbrace{\sum_i (y^{(i)} - \hat{y}^{(i)})^2}_{\text{SSE}}$$

$$1 = \underbrace{\frac{SSM}{SST}}_{r^2} + \underbrace{\frac{SSE}{SST}}_{1-r^2}$$

- r^2 is called the coefficient of determination (square of coefficient of correlation!)
- Captures the fraction of variability explained by model
- It is a measure that allows us to determine how certain one can be in making predictions from a certain model/graph
- closer to 1, the better.

Polynomial Fit